Phosphorus is a nonmetallic element that plays an important role in biological systems as a component of ATP and the sugar phosphate backbone of DNA. Elemental phosphorus can exist in several different structural forms, known as allotropes, each exhibiting different chemical properties. The two most common allotropes of phosphorus are P_4 (white phosphorus) and $(P_4)_n$ (red phosphorus).

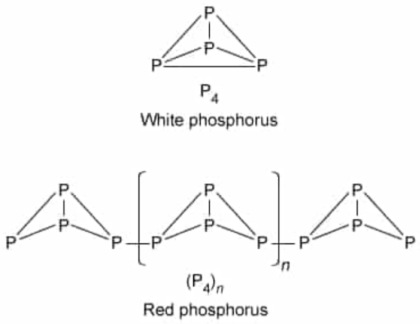


Figure 1 Structural representation of the P_4 and $(P_4)_n$ allotropes of phosphorus

P_4 is thermodynamically unstable due to its constrained geometric structure and ignites upon exposure to air. However, the polymeric structure of $(P_4)_n$ results in greater thermodynamic stability under ambient conditions. Upon exposure to intense heat at pressures less than 43 atm, $(P_4)_n$ sublimes and decomposes into P_4.

In an experiment, a sealed silica tube containing a sample of solid $(P_4)_n$ at one end was placed under 1 atm of inert gas. The sample was continuously heated at 450 °C until all the $(P_4)_n$ had sublimed. The resulting vapors were transferred by the inert gas to the other cooled end of the tube, where the vapors condensed into liquid P_4. Further cooling solidified the P_4 product, and a 5.74 g sample was collected from the tube. The heating curve shown in Figure 2 was subsequently obtained for the P_4 sample.

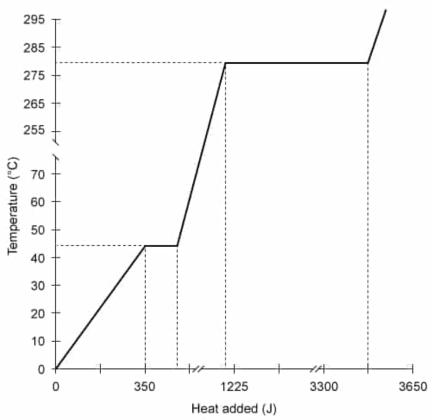


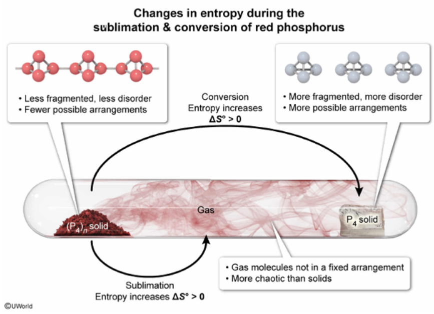
Figure 2 Heating curve of white phosphorus at a pressure of 1 atm (Note: Heat was supplied at a constant rate during the experiment, and the latent heat of fusion ΔH_{fusion} was measured to be 2.51 kJ/mol.)

Which of the following describes the standard change in entropy ΔS for the sublimation of $(P_4)_n(s)$ and the overall conversion of $(P_4)_n(s)$ to $P_4(s)$, respectively?

- A. $\Delta S_{sublimation} > 0$ and $\Delta S_{conversion} < 0$ [39%]
- B. $\Delta S_{sublimation} < 0$ and $\Delta S_{conversion} < 0$ [5%]
- ✔ C. $\Delta S_{sublimation} > 0$ and $\Delta S_{conversion} > 0$ [44%]
- D. $\Delta S_{sublimation} < 0$ and $\Delta S_{conversion} > 0$ [9%]

Correct
43% answered correctly

Explanation:



Entropy is a measure of the molecular disorder of a system. Solids contain atoms that are closely packed in a fixed arrangement with little room to move, resulting in relatively low entropy. Conversely, gases have relatively high entropy because they are spaced far apart and are free to move into many different arrangements. The entropy of liquids is between that of solids and gases ($S_{solid} < S_{liquid} < S_{gas}$).

During a chemical reaction, the total entropy of a system changes, and the standard change in entropy $\Delta S°$ is evaluated as:

$$\Delta S = \Sigma S_{products} - \Sigma S_{reactants}$$

A reaction that produces a less ordered (ie, more random) system yields a positive ΔS, whereas a reaction that produces a more ordered system yields a negative ΔS.

Consider the change in entropy for the following processes involving elemental phosphorus:

- **Sublimation of $(P_4)_n$** is the direct conversion of $(P_4)_n(s)$ to $(P_4)_n(g)$. This process increases the total entropy of the system (ie, 1 mole of solid converts to 1 mole of gas), resulting in **$\Delta S > 0$**.
- **Conversion of $(P_4)_n(s)$ to $P_4(s)$** breaks up the polymer chains of repeating P_4 monomer units. This results in less restricted movement of the P_4 units (ie, the number of possible arrangements increases) and yields more disorder (higher entropy) in $P_4(s)$ compared to the layered polymer chains of $(P_4)_n(s)$. Therefore, the conversion of $(P_4)_n(s)$ to $P_4(s)$ yields **$\Delta S > 0$**.

(Choices A, B, and D) Both sublimation of $(P_4)_n(s)$ and conversion of $(P_4)_n(s)$ to $P_4(s)$ produce an *increase* in entropy ($\Delta S > 0$).

Educational objective:
Entropy describes the disorder of a system. Gases are disordered (high entropy), whereas solids are highly ordered (low entropy). Monomers are less ordered and have a higher entropy than layered polymers.

Time spent: 0 sec
Subject: General Chemistry

QID: 403312
System: Thermodynamics, Kinetics & Gas Laws

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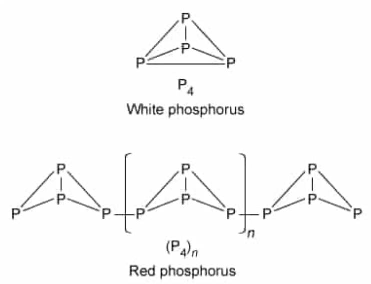

Figure 1 Structural representation of the P_4 and $(P_4)_n$ allotropes of phosphorus

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In an experiment, a sealed silica tube containing a sample of solid $(P_4)_n$ at one end was placed under 1 atm of inert gas. The sample was continuously heated at 450 °C until all the $(P_4)_n$ had sublimed. The resulting vapors were transferred by the inert gas to the other cooled end of the tube, where the vapors condensed into liquid P_4. Further cooling solidified the P_4 product, and a 5.74 g sample was collected from the tube. The heating curve shown in Figure 2 was subsequently obtained for the P_4 sample.

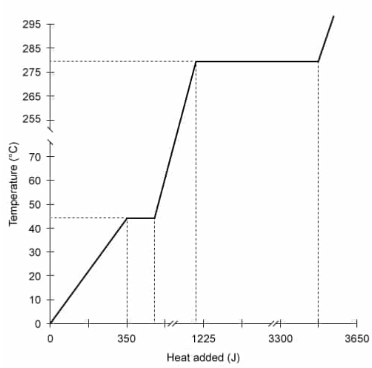


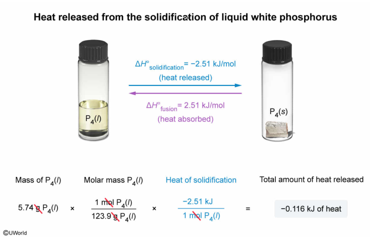
Figure 2 Heating curve of white phosphorus at a pressure of 1 atm (Note: Heat was supplied at a constant rate during the experiment, and the latent heat of fusion ΔH_{fusion} was measured to be 2.51 kJ/mol.)

Approximately how much energy in the form of heat is released when 5.74 g of $P_4(l)$ is solidified?

- A. 0.0184 kJ [5%]
- B. 0.116 kJ [55%]
- C. 2.51 kJ [25%]
- D. 14.4 kJ [12%]

Correct
54% answered correctly

Explanation:



When a solid substance is heated to its melting point, additional heat energy must be **absorbed** by the system to break the attractive intermolecular forces between the solid molecules and enable the energized solid molecules to break apart and enter the liquid phase. This absorbed heat energy is called the **heat of fusion ΔH_{fusion}** and is measured as the amount of energy that must be absorbed to convert 1 mole of solid to the liquid phase. Because this additional heat is absorbed by the substance, ΔH_{fusion} does not contribute to an increase in temperature and is seen as a horizontal segment on a heating curve.

Alternatively, when a liquid substance is cooled to its freezing point, an additional amount of heat energy equal to ΔH_{fusion} must be **released** as the liquid is converted into the solid phase. As such, the energy transferred for complimentary phase changes (ie, freezing and melting) is equal in magnitude but opposite in sign (ie, a positive sign indicates energy absorbed, and a negative sign indicates energy released). Consequently,

$$-\Delta H_{\text{fusion}} = \Delta H_{\text{solidification}}$$

In this question, 5.74 g of P_4 liquid is converted to the solid phase. Dividing the mass of $P_4(l)$ by its molar mass, which is found using the periodic table, yields the number of moles of $P_4(l)$:

$$5.74\ \text{g}\,\text{P}_4(l) \times \frac{1\ \text{mol}\ \text{P}_4(l)}{123.9\ \text{g}\,\text{P}_4(l)} = 0.0463\ \text{mol}\ \text{P}_4(l)$$

The passage states that the heat of fusion for P_4 is 2.51 kJ/mol. Multiplying the moles of $P_4(l)$ by $-\Delta H_{\text{fusion}}$ gives the amount of heat:

$$0.0463\ \cancel{\text{mol}}\,\text{P}_4(l) \times \frac{-2.51\ \text{kJ}}{1\ \cancel{\text{mol}}\ \text{P}_4(l)} = -0.116\ \text{kJ}$$

The negative sign indicates that heat is released. Therefore, 0.116 kJ of heat are released during the phase change.

(Choice A) The amount of 0.0184 kJ is calculated if moles of $P_4(l)$ are *divided* by ΔH_{fusion}, rather than *multiplied*.

(Choice C) 2.51 kJ is the amount of heat released when *1 mole* of $P_4(l)$ is converted to $P_4(s)$.

(Choice D) A quantity of 14.4 kJ is obtained if the mass of $P_4(l)$ is *not* converted to moles.

Educational objective:
The heat of fusion is defined as the amount of heat needed to convert 1 mole of solid to the liquid phase. For complimentary phase changes, the amount of heat transferred during the process is equal in magnitude but opposite in sign.

Time spent: 0 sec
Subject: General Chemistry

QID: 403313
System: Thermodynamics, Kinetics & Gas Laws

Phosphorus is a nonmetallic element that plays an important role in biological systems as a component of ATP and the sugar phosphate backbone of DNA. Elemental phosphorus can exist in several different structural forms, known as allotropes, each exhibiting different chemical properties. The two most common allotropes of phosphorus are P_4 (white phosphorus) and $(P_4)_n$ (red phosphorus).

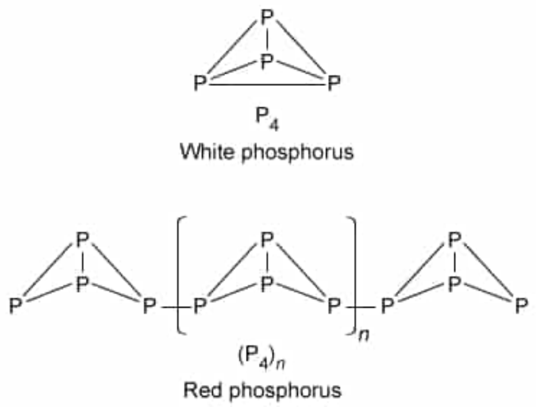


Figure 1 Structural representation of the P_4 and $(P_4)_n$ allotropes of phosphorus

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In an experiment, a sealed silica tube containing a sample of solid $(P_4)_n$ at one end was placed under 1 atm of inert gas. The sample was continuously heated at 450 °C until all the $(P_4)_n$ had sublimed. The resulting vapors were transferred by the inert gas to the other cooled end of the tube, where the vapors condensed into liquid P_4. Further cooling solidified the P_4 product, and a 5.74 g sample was collected from the tube. The heating curve shown in Figure 2 was subsequently obtained for the P_4 sample.

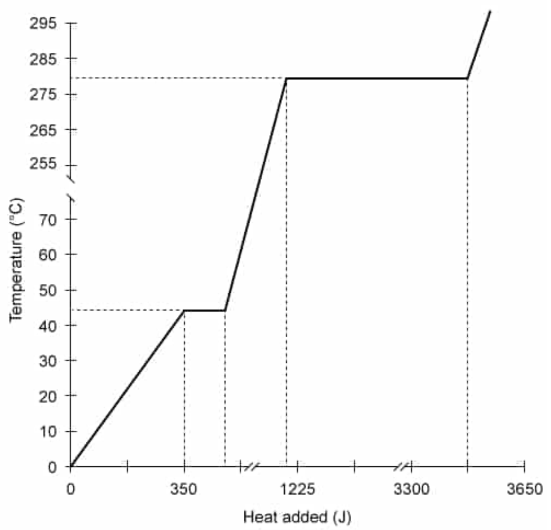


Figure 2 Heating curve of white phosphorus at a pressure of 1 atm (Note: Heat was supplied at a constant rate during the experiment, and the latent heat of fusion ΔH_{fusion} was measured

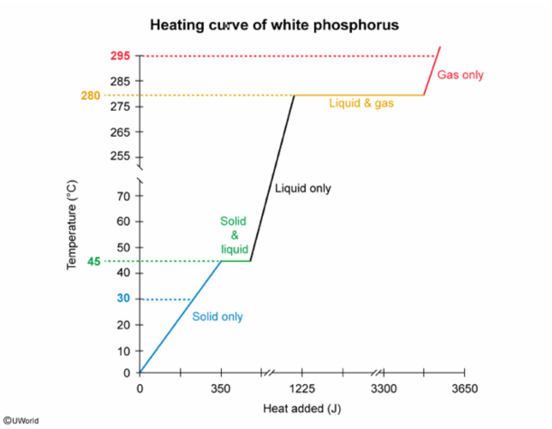
to be 2.51 kJ/mol.)

Which of the following does NOT describe the phase behavior of P_4 at 1 atm of pressure?

- A. At 30 °C, P_4 exists as a solid. [3%]
- B. At 45 °C, P_4 exists as a mixture of a solid and a liquid. [4%]
- ✔ C. At 280 °C, P_4 exists as a mixture of a solid, liquid, and gas. [86%]
- D. At 295 °C, P_4 exists as a gas. [4%]

Correct
87% answered correctly

Explanation:



A **heating curve** graphically represents how the temperature of a substance changes as heat is absorbed at a constant rate. The plateaus (ie, the horizontal flat segments) of a heating curve represent phase changes (eg, melting or boiling). Along these segments, added heat does not result in a temperature increase. Rather, heat is used to overcome the attractive intermolecular forces holding the melting or boiling molecules together. After the substance has fully changed phase, any additional heat contributes to increasing the temperature.

According to the heating curve for P_4:

- Over the range of **0 °C to 45 °C**, the temperature of solid P_4 (ie, the lowest-energy phase) **increases linearly** with added heat, indicating that no phase change is taking place. Accordingly, P_4 exists **only as a solid at 30°C (Choice A)**.

- At **45 °C**, the first phase change occurs (indicated by the first plateau). As the solid absorbs more heat, it begins to melt while the temperature (ie, the melting point temperature) **stays constant**. Both **solid and liquid** phases are present over the length of this segment **(Choice B)**.
- At **280 °C**, the second phase change occurs (indicated by second plateau). As the liquid absorbs more heat, it begins to boil while the temperature (ie, the boiling point) **stays constant**. Both **liquid and gas** phases are present over the length of this segment.

- Over the range of **280 °C to 300 °C**, the temperature of gaseous P_4 (ie, the highest-energy phase) **increases linearly** with added heat, indicating that no phase change is taking place. As such, P_4 exists **only as a gas at 295 °C (Choice D)**.

Therefore, **Choice C** does NOT describe the phase behavior of P_4 at 280 °C.

Educational objective:
A heating curve graphically shows how the temperature of a substance changes with the addition of heat. Flat (ie, constant temperature) regions of the heating curve represent phase changes.

Time spent: 0 sec
Subject: General Chemistry
QID: 403314
System: Thermodynamics, Kinetics & Gas Laws

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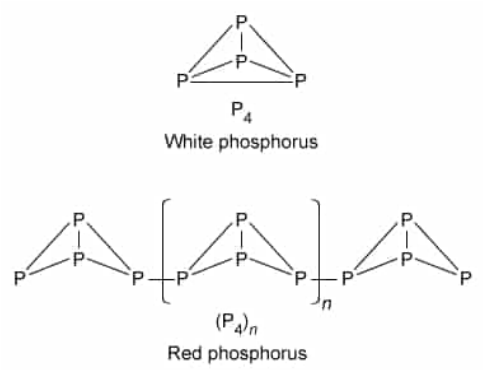

Figure 1 Structural representation of the P_4 and $(P_4)_n$ allotropes of phosphorus

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In an experiment, a sealed silica tube containing a sample of solid $(P_4)_n$ at one end was placed under 1 atm of inert gas. The sample was continuously heated at 450 °C until all the $(P_4)_n$ had sublimed. The resulting vapors were transferred by the inert gas to the other cooled end of the tube, where the vapors condensed into liquid P_4. Further cooling solidified the P_4 product, and a 5.74 g sample was collected from the tube. The heating curve shown in Figure 2 was subsequently obtained for the P_4 sample.

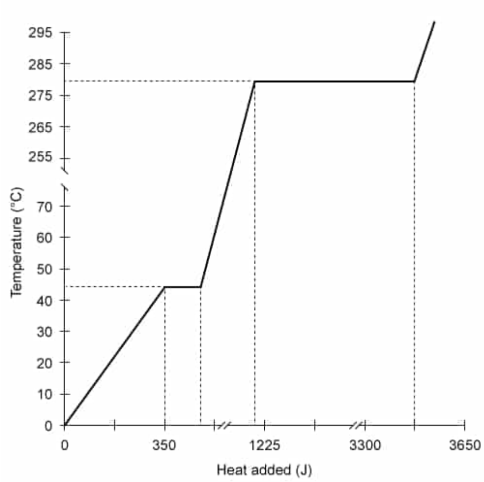

Figure 2 Heating curve of white phosphorus at a pressure of 1 atm (Note: Heat was supplied at a constant rate during the experiment, and the latent heat of fusion ΔH_{fusion} was measured to be 2.51 kJ/mol.)

The magnitude of the latent heat of vaporization $\Delta H_{vaporization}$ for P_4 is:

- ✔ ◉ A. greater than ΔH_{fusion}. [81%]
- ○ B. equal to ΔH_{fusion}. [9%]

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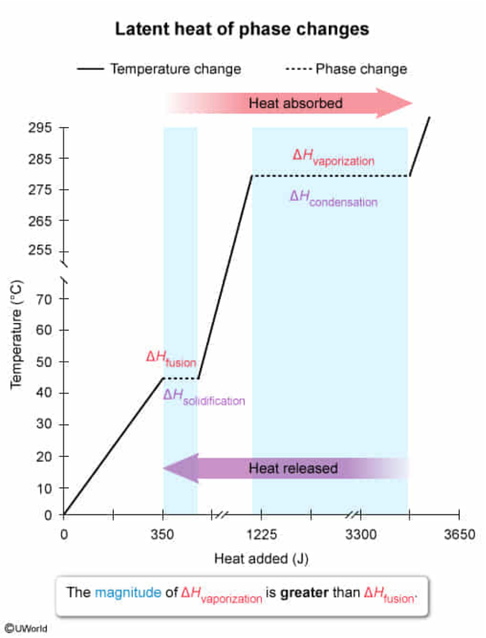
C. less than $\Delta H_{condensation}$. [7%]

D. equal to $\Delta H_{solidification}$. [2%]

Correct
81% answered correctly

Explanation:



When enough energy (transferred as heat) is either absorbed or released by a substance, a phase change can occur. During this process, the temperature of the substance remains constant until the substance has completely transformed from one phase to another. The **latent heat** is the amount of heat that 1 mole of a substance must absorb or release to undergo a phase change at constant temperature. The latent heat transferred for complimentary phase changes (eg, freezing and melting) is **equal in magnitude but opposite in sign** (ie, the sign indicates if the heat is absorbed or released).

There are two common types of latent heat:

- **Latent heat of fusion (ΔH_{fusion})** is the amount of heat a substance must **absorb** to transition from a solid to a liquid (ie, melting). The opposite process (ie, going from liquid to a solid) **releases** energy and the latent heat **$\Delta H_{solidification}$** of this reverse process is equal to $-\Delta H_{fusion}$.
- **Latent heat of vaporization ($\Delta H_{vaporization}$)** is the amount of heat a substance must **absorb** to transition from a liquid to a gas (ie, boiling). The opposite process (ie, going from gas to a liquid) **releases** energy, and the latent heat **$\Delta H_{condensation}$** for this reverse process is equal to $-\Delta H_{vaporization}$.

A phase change is indicated on a heating curve as a plateau (ie, flat region). In the given heating curve for P_4, the length of the first plateau (ie, the lowest-temperature phase change) represents the magnitude of ΔH_{fusion}, and the length of the second plateau (ie, the highest-temperature phase change) represents the magnitude of $\Delta H_{vaporization}$. Because the length of the second plateau is greater than that of the first, the magnitude of $\Delta H_{vaporization}$ is greater than ΔH_{fusion}.

(Choices B and D) Because the magnitude of $\Delta H_{vaporization}$ is *greater* than ΔH_{fusion}, its magnitude is also *greater* than that of $\Delta H_{solidification}$ (ie, fusion and solidification are complimentary phase changes).

(Choice C) Vaporization and condensation are complimentary phase changes; consequently, $\Delta H_{vaporization}$

and $\Delta H_{condensation}$ are *equal* in magnitude.

Educational objective:

Latent heat is the amount of heat 1 mole of a substance must absorb (or release) to undergo a phase change at constant temperature. A phase change is indicated on a heating curve as a plateau; the length of the plateau represents the magnitude of the latent heat of the transition.

Time spent:0 sec
Subject:
General Chemistry

QID:403315
System:
Thermodynamics, Kinetics & Gas Laws

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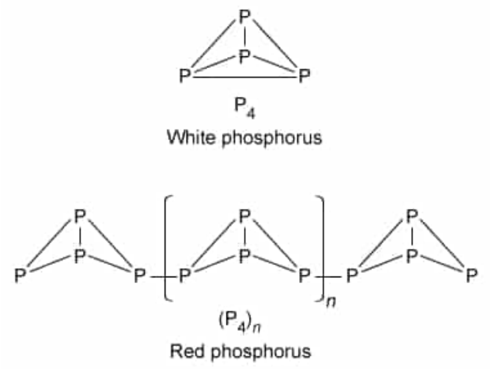

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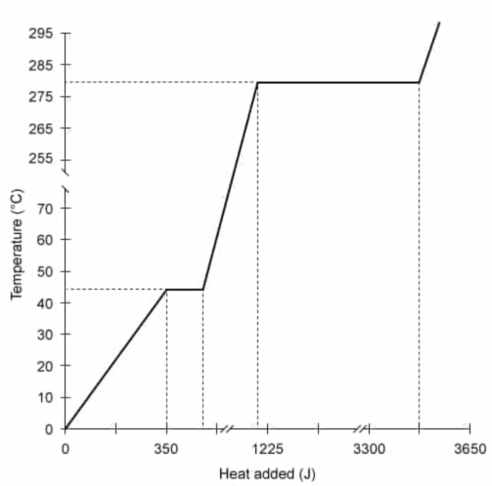


Figure 2 Heating curve of white phosphorus at a pressure of 1 atm (Note: Heat was supplied at a constant rate during the experiment, and the latent heat of fusion ΔH_{fusion} was measured to be 2.51 kJ/mol.)

The phosphorus allotrope that has the highest melting point is:

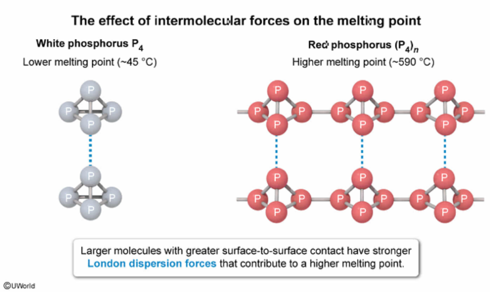
- ✔ ◉ A. $(P_4)_n$, because it has the stronger intermolecular forces. [58%]
- ○ B. P_4, because it has the stronger intermolecular forces. [7%]

◯ C. $(P_4)_n$, because it has the stronger P–P bonds that require more energy to break. [31%]

◯ D. P_4, because it has the stronger P–P bonds that are much more polar. [2%]

Correct
58% answered correctly

Explanation:



The **melting point** is the temperature at which a substance in the solid-state transitions to the liquid state. At the melting point, both solid and liquid states coexist in equilibrium. Melting occurs when the kinetic energy of a molecule is high enough to overcome the attractive intermolecular forces present within the solid. Substances with **stronger intermolecular forces** have correspondingly **higher melting points**.

London dispersion forces, a type of intermolecular force, occur when two neighboring, nonpolar species induce a temporary distortion of the electron clouds, resulting in weak, momentary dipoles. The strength of the attractive force between the dipoles depends on the size and shape of the species. Larger molecules have larger electron clouds that are more polarizable and yield stronger attractive London forces. Furthermore, the shape of the molecule affects the amount of polarizable orbital "surface area" available to interact with surrounding molecules. The greater the surface-to-surface contact between molecules, the greater the London dispersion forces.

Both P_4 and $(P_4)_n$ are nonpolar molecules that experience London dispersion forces. However, $(P_4)_n$ comprises repeating P_4 monomer units linked together, resulting in a larger molecular size and orbital surface area. Therefore, $(P_4)_n$ will have a higher melting point than P_4 due to its stronger London dispersion forces.

(Choice B) Because $(P_4)_n$ experiences stronger intermolecular forces due to its increased surface area and size, it will have a higher melting point than P_4.

(Choice C) The P–P bonds have essentially the same strength in both compounds (ie, require about the same energy to break), but this is irrelevant with regard to the melting point. Melting occurs when enough energy is gained to overcome noncovalent intermolecular forces. The P–P bonds are covalent bonds, which are not broken during melting.

(Choice D) The atoms in the P–P bonds are the same and have no difference in electronegativity (ie, the bonds are completely nonpolar). Bond polarity cannot account for the difference in the melting points between the two compounds.

Educational objective:
Molecules with larger, more polarizable electron clouds have stronger London dispersion forces and higher melting points because more energy is required to overcome the enhanced attraction between neighboring molecules.

Time spent:0 sec **QID:**403316

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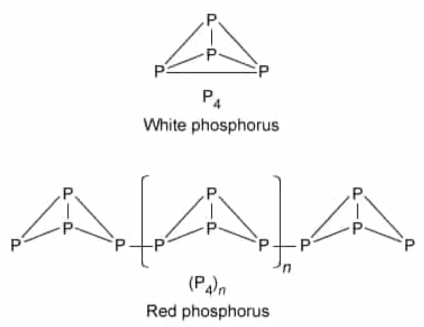

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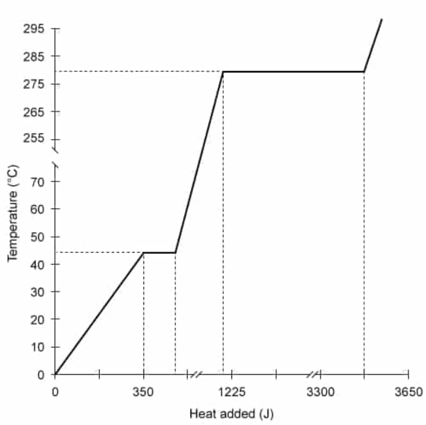


Figure 2 Heating curve of white phosphorus at a pressure of 1 atm (Note: Heat was supplied at a constant rate during the experiment, and the latent heat of fusion ΔH_{fusion} was measured to be 2.51 kJ/mol.)

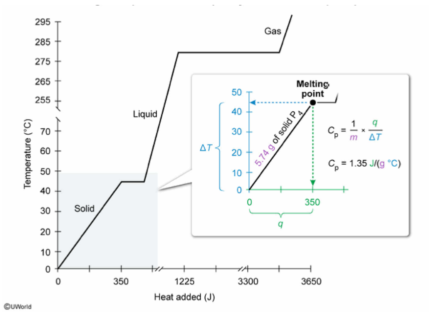
What is the specific heat capacity of solid P_4?

- A. 0.0224 J/(g·°C) [6%]
- B. 0.128 J/(g·°C) [15%]
- ✔ C. 1.35 J/(g·°C) [68%]
- D. 7.78 J/(g·°C) [8%]

Correct
67% answered correctly

Explanation:

Calculating the specific heat capacity of solid white phosphorus



The specific heat capacity C_p of a substance (commonly reported in units of J/g·°C) is defined as the amount of heat (q) required to raise the temperature of **1 gram** of the substance by **1 °C**. In other words, the specific heat capacity describes the extent to which 1 gram of a substance will change temperature upon exposure to a given amount of heat.

The C_p of a substance can be determined using the following expression for q:

$$q = mC_p \, \Delta T = mC_p(T_{\text{final}} - T_{\text{initial}})$$

where m as the mass of the substance and ΔT is the change in temperature. Rearranging the expression to solve for C_p yields

$$C_p = \frac{q}{m \, \Delta T}$$

According to the heating curve obtained using the 5.74 g sample of $P_4(s)$, a total of 350 J of heat (x-axis) is required to raise the temperature (y-axis) of solid P_4 from 0 °C (T_{initial}) to its melting point of 45 °C (T_{final}). By substituting these values into the equation, the specific heat capacity of solid P_4 can be calculated:

$$C_p = \frac{q}{m(T_{\text{final}} - T_{\text{initial}})} = \frac{350\ \text{J}}{5.74\ \text{g} \times (45°\text{C} - 0°\text{C})} = \frac{350\ \text{J}}{258.3\ \text{g} \cdot °\text{C}} = 1.35 \frac{\text{J}}{\text{g} \cdot °\text{C}}$$

Note that choosing any (x,y) coordinate along the heating curve where P_4 exists as a solid will result in C_p = 1.35 J/(g·°C).

(Choice A) 0.0224 J/(g·°C) results from incorrectly evaluating the specific heat capacity as $C_p = \frac{\Delta T}{\text{mq}}$ rather than $C_p = \frac{q}{m \, \Delta T}$.

(Choice B) 0.128 J/(g·°C) is obtained if ΔT is divided by q (ie, the *slope* of the segment of the heating curve for solid P_4).

(Choice D) 7.78 J/(g·°C) results from dividing q *only* by ΔT, which does not account for the mass of the sample and yields the heat capacity of the sample, not the *specific* heat capacity (ie, heat capacity = mass × specific heat capacity).

Educational objective:
The specific heat capacity C_p of a substance is the amount of heat required to raise the temperature of 1 gram of the substance by 1 °C.

Time spent:0 sec
Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws